MUSLIMHACKS 2026 · ELDERLY CARE · CONCORDIA UNIVERSITY

Amanah Care

The care record only your family can read.

Abdul Quayum Adekunle · safio

amanah = a trust you are responsible for

Ammi is 78. Three of her children share her care. Here is the system.

Fatima · 07:52
did she take the morning ones?

Sister A · 08:10
yes. she ate half. who has her this afternoon?

Abdullah · 08:41
I thought Fatima. I have the pharmacy at 11 I think?

Fatima · 09:15
no that was last week. what did Dr Rahman say wednesday??

Sister A · 09:20
scroll up 🙃

- No plan, so things get missed or done twice.
- No handoff, so the next person does not know what happened.
- No record of who did what, so one sibling carries it all.
- And the whole history of her health sits with a chat company.

CARE IS ALREADY HAPPENING. IT IS JUST NOT TRACKED.

1 in 4

American Muslims care for an older adult, versus 16% of the general public.

ISPU, Support and Sacrifice, 2026 (American Muslim Poll 2025)

52%

of Muslims aged 30 to 49 care for an elder. 21% of 18 to 29, 31% of 50 and over.

ISPU, Sandwiched

42%

of Canadians 15 and over gave unpaid care in 2022: 13.4 million people, 1.8 million of them sandwiched.

Statistics Canada, Canadian Social Survey 2022

31% depression, 49% burden among family caregivers of people with dementia, across 43 studies.

Collins & Kishita, Ageing & Society, 2020, meta-analysis

18% vs 7% of Muslim caregivers report difficulty finding culturally appropriate care.

ISPU 2026, cited in the MuslimHacks 2026 Challenge Guide

"Medication, appointments, meals, transport and personal care may be tracked across WhatsApp, memory and separate calendars." Challenge Guide. Health information needs consent and privacy protection under Quebec Law 25, Ontario PHIPA and PIPEDA. A chat group meets none of that.

ROOT CAUSE

WhatsApp is a conversation. Care needs a record.

A record has five parts. Existing tools each do one. None do all five with the elder's preferences built in, and none keep the family in control of the data.

Plan

Her routine: meds, meals, prayers, walk, pickups.

Actuals

What was really done, by whom, when.

Handoff

What happened. What is next.

Load

Who carried the week.

Privacy

Only the family can read it.

WHAT WE BUILT

Amanah Care: a private care record for families.

The next caregiver always knows what happened and what is next. The family sees who did what. Nobody else can read a word of it.

Routine	Ammi's plan, once. Every day becomes a checklist measured against it.
Log	Two taps: category, what happened. A note if you want.
Hand off	Summary written for you from the log. "What is next" written for you from the routine.
Dashboard	Who has Ammi right now. Meds 2 of 3. Mood, last logged. Next pickup and who is on it. Who carried the week.
Family key	Made on the phone, shared by QR in person. The server stores scrambled text and nothing else.

LIVE DEMO

One afternoon with Ammi's family, on two phones.

1. **Sister A** opens Home: Ammi's day, 8 of 12 done, meds 2 of 3, next pickup Sunday 11:00, Abdullah.
2. She taps **Done** on dinner. The plan and the tiles update.
3. She opens **Hand off** to Fatima. What happened and what is next are already written. She sends.
4. **Fatima's phone** lights up. She reads the card with Ammi's preferences on top, and accepts.
5. Home now says **Fatima has Ammi**. The week's load bar moves.
6. We open the **database**. Every row is scrambled text.

What the judges should watch for

- Nothing typed except one line, everything else composed from the record.
- Two real browsers, one real backend, live handoff in seconds.
- The same data, readable on the phones, unreadable on the server.

WHY THIS BEATS THE FAMILY GROUP CHAT

QUESTION A CAREGIVER ASKS	WHATSAPP GROUP	AMANAH CARE
What is Ammi's plan today?	In someone's head	**Today's plan, 8 of 12 done**
Did she take her afternoon meds?	Scroll and ask	**Meds 2 of 3, ✓ Sister A 14:05**
What do I need to know for tonight?	Whatever got typed	**Handoff card: what happened, what is next**
Who is taking her to the pharmacy?	"I thought you were"	**Sunday 11:00, Abdullah, on the dashboard**
Is one of us doing everything?	Invisible until someone burns out	**Who carried the week, per person**
Her language, diet, prayer, modesty?	Retold every time	**On top of every handoff card**
Who can read her health history?	Meta, and anyone forwarded	**Only phones holding the family key**

HOW IT WORKS

Every action is an event. Every event is sealed before it leaves the phone.

1

Phone

Plaintext + family key H. Encrypts with AES-GCM, fresh IV each time.

READABLE

2

API

Node. Checks membership, appends. Holds no key.

SEALED

3

Redis Streams

Append-only event log. Replayable.

SEALED

4

Projector

Consumer group. Idempotent by event id.

SEALED

5

Postgres

4 tables + 1 view. Read models from metadata only.

SEALED

6

Family phones

Poll, decrypt, compute the dashboard locally.

READABLE

Why event sourcing

The dashboard, the routine, the record and the workload were added without touching the write path. They are projections over the same log. Replay rebuilds everything.

Runs in one command

Docker Compose: web, api, projector, redis, postgres. Fallback: one node process runs the same api and projector on an in-memory Postgres.

PRIVACY BY DESIGN, NOT BY POLICY

Even we, the operators, cannot read Ammi's day.

- The family key is made in the browser. It is never sent anywhere.
- The server keeps a SHA-256 check of the key: enough to prove you hold it, never to read with it.
- Every payload is AES-GCM with a fresh 96-bit IV. Tampering fails to decrypt.
- Members join by QR, face to face. The app never offers "share to WhatsApp".
- Clear on the server: event type, category, who, when. Sealed: names, notes, summaries, preferences, routine.
- Workload is counted from metadata. No decryption needed.
- No LLM anywhere in the loop. The server could not run one on this data if it wanted to.
- A test asserts a logged note appears in no server-side row, no stream entry, no API response.

TESTED

40 automated tests. Real routes, real schema, in process.

100%

line coverage

api/app.js · projector/apply.js · web/crypto.js

90%

branch coverage

Node's built-in test runner and coverage. No mocks of our own code.

100%

function coverage

Every exported function exercised.

What the suite checks

- Guards: non-members rejected, only the elder may change consent.
- Stream fields, idempotent replay, handoff lifecycle open → acknowledged.
- AES-GCM round trip, fresh IV per message, wrong key and tampering give null.

How it runs

The real `init.sql` loads into an in-memory Postgres. Requests hit the real Express app. A fake stream records every XADD and pushes it through the real projector before the read models are checked. `npm test`, 2.5 seconds, no Docker.

WHAT IS REAL, WHAT IS SIMPLIFIED, WHAT IS NOT BUILT

REAL

- Create family, join by QR or code
- Log, routine, today's plan
- Handoff: compose, send, accept
- Dashboard, record, workload
- End-to-end encryption, ciphertext-only server
- Redis Streams → projector → Postgres
- Test suite with coverage

SIMPLIFIED

- Notifications by polling every 4 s, not push
- Identity is a trusted member id, no login yet
- One Redis stream for all families, one consumer group
- Support workers hold the same key as family

NOT BUILT

- Elder consent filter (needs a scoped support key)
- Key rotation after a lost phone
- Reminders on open handoffs
- Elder large-print view, Urdu and Arabic
- Native mobile apps

We chose one workflow, the handoff, and built it properly. Everything on the right is designed and on the roadmap, not hidden.

WHO PAYS, AND WHAT IT COSTS TO RUN

Families: free

The record is theirs. Free keeps the WhatsApp-replacement honest and adoption in-person, family to family, masjid to masjid.

Agencies and community care: paid

Home-care agencies and community programs that place support workers need handoff logs and scoped access. Per elder, per month. The support role already exists in the model.

Cost of running: small

Stateless API, a stream, a database. The server holds ciphertext only, so it is not a health-data honeypot. One small VM serves thousands of families.

How we know it works

Plan adherence per day. Handoffs accepted within 30 minutes. Number of active caregivers per family, because the point is spreading the load. Weekly return rate.

WHAT IS NEXT

Monday

- Five Montreal families, Urdu and Arabic speaking, one month pilot
- Elder large-print "today" view for the tablet on the kitchen table

Next month

- Scoped key for support workers, so the consent filter is cryptographic, not cosmetic
- Push notifications, key rotation, real login

Later

- Native apps, offline first
- Voice to structured log, on the phone, opt in per family. AI at the edge, never on the server
- Per-family streams, multi-region

WHERE AI FITS

On the phone, where the plaintext already is. Never on the server.

Say it, do not type it

"Ammi ate half her lunch, took her meds, seemed tired" becomes three log items: meal, meds, mood. The log already takes structured items, so a parser plugs in front of it.

Read it her way

Handoff summary in plain sentences, and in Urdu or Arabic for the elder's large-print view. Translation on the device.

Patterns, not predictions

"Refused lunch 3 of the last 5 days." Counted from the record, shown as a fact. The family decides what it means. No diagnosis, no triage.

How, without breaking the privacy claim

Chrome's built-in on-device APIs (Prompt, Summarizer, Translator on Gemini Nano, stable on desktop since Chrome 148) and Apple's on-device models in the native app. Opt in per family. Nothing leaves the phone.

Why not tonight

A cloud model call from the demo would contradict "the server cannot read it" on stage. So it is roadmap, designed to fit the architecture rather than bolted on.

REMEMBER THIS

Care was already happening.
Now it has a record.
And only the family can read it.

We ask Introductions to community care programs and five families for the pilot.

Team Abdul Quayum Adekunle · safio

Delivery format?	Web app, phone first, any browser. Two windows on stage stand in for two phones. Native mobile is roadmap.
Technologies?	Node/Express API, Redis Streams event log, Postgres read models, vanilla JS + WebCrypto AES-GCM, nginx, Docker Compose. Node's built-in test runner. Vendored QR library. No frameworks, no CDNs, so the demo needs no internet.
Behind the scenes?	Phone encrypts → POST /events → membership check → XADD to the stream → projector consumer group → Postgres events + handoffs + workload view → phones poll and decrypt. The dashboard is computed on the phone.
Built from scratch here?	Spec, requirements and two audit rounds were planned ahead. All code was written during the hackathon window. QR generation is a vendored library.
Tradeoffs?	Polling over push, trusted member id over login, one stream over per-family streams, consent filter deferred. Each one on the honesty slide with the reason.

Target user?	Adult children sharing care of an elderly parent. The elder is the subject and owner of the preferences, not the app user. Support workers join with their own role.
Discovery and adoption?	Family to family, by QR, in person, by design. Community care programs and masjid welfare committees as the channel. Agencies bring their families.
How many benefit?	1 in 4 American Muslims already care for an older adult (ISPU 2026). Muslims are 8.7% of Greater Montreal (Census 2021). Start with Montreal families, then any family with a shared care load.
Impact if fully developed?	Fewer missed meds and appointments, a fairer split of the load, and less of the strain ISPU documents. A family-owned health record the family never had.
Success measure?	Plan adherence, handoff acceptance time, active caregivers per family, weekly return rate.

Fully functional or mocked?	The core loop is real end to end: encrypt, stream, project, store, decrypt. Notifications are polling. Identity is stubbed. The consent filter is not built. Nothing on stage is faked.
Most important feature?	The handoff, composed from the record: what happened from the log, what is next from the routine, the elder's preferences on top.
Hardest part?	Keeping the server useful while it can read nothing: deciding which fields stay clear for routing and counting, and building a whole dashboard from client-side decryption.
Business model?	Free for families. Paid per elder per month by home-care agencies and community programs that need handoff logs and scoped access.
Scale beyond the prototype?	Per-family streams, push, scoped keys, real login, offline. Ciphertext-only storage keeps compliance and hosting light.
Who pays?	Agencies and community organisations. Families optionally for extras like key escrow between two relatives.

How did you divide the work?	Abdul: architecture, event model, crypto, api and projector, dashboard, tests. safio: in-app invite flow, interface cleanup, the flow view that traces a write hop by hop.
What did you learn?	Cut scope early and say so. One workflow done well beats five half done. Privacy shapes every feature, including which ones you cannot build yet.
Differently next time?	Tests from hour one. Push notifications budgeted from the start.
How did you prioritise?	Requirement register of 46 items, two audit rounds on feasibility in 24 hours, then "cut core": the handoff loop first, everything else labelled roadmap.
What data?	Care events, handoff summaries, preferences, routine, member names, all encrypted. Clear: event type, category, member id, timestamps.
Ethical risks?	Support workers currently hold the same key, so hiding categories is UI only. Lost key means lost data until rotation ships. No diagnosis, dosage or triage anywhere, and the metrics are counts, never judgments.
Depends on an LLM?	No. By design the server cannot run one on this data.

Next if more time?	Elder large-print view in Urdu and Arabic, scoped support key, push, key rotation.
Continue after today?	Yes. Five-family pilot in Montreal next week.
First step after today?	Commit and publish the repo, deploy on one small VM, recruit the pilot families through community care contacts.
Different from existing options?	Task apps organise chores, med apps remind, marketplaces find carers. None connect plan, actuals, handoff and load with the elder's preferences, and none let only the family read the data.
Help needed?	Introductions to community care programs and agencies, pilot families, and a designer for the elder view.
Remember most?	The server cannot read it. Open the database and see for yourself.
Why should this win?	It solves the coordination problem the track asked for, with the elder's preferences built in, privacy that is cryptographic rather than a promise, an honest scope, and a test suite behind it.

APPENDIX · SOURCES

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